

Lemma

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① Conditioning over an event A

$$f_{X|A}(x) = P(\{X=x\} \cap A)$$

Suppose $A = \{X \in B\}$

$$f_{X|A}(x) = \begin{cases} \frac{f_X(x)}{P(X \in B)} & x \in B \\ 0 & \text{otherwise} \end{cases}$$

② Total probability theorem

$$f_X(x) = \sum_{i=1}^n f_{X|A_i}(x) P(A_i)$$

Suppose Y is a discrete RV; $A_y = \{Y=y\}$ then $f_X(x) = \sum_{y \in \mathcal{Y}} f_{X|Y=y}(x) P_Y(y)$

$f_{X|Y=y}(x)$ is a function

$$\begin{aligned} \text{st} \quad F_{X|Y=y}(x) &= \int_{-\infty}^x f_{X|Y=y}(u) du \\ &= P(X \leq x | Y=y) \\ &= F_{X|Y}(x|y) \end{aligned}$$

X continuous and Y discrete

$$F_{X|Y}(x|y) = P(X \leq x | Y=y) = \frac{P(X \leq x, Y=y)}{P(Y=y)}$$

and $f_{X|Y}(x|y)$ is pdf such that

$$F_{X|Y}(x|y) = \int_{-\infty}^x f_{X|Y}(u|y) du$$

X continuous & Y is continuous.

$$F_{X|Y}(x|y) = \lim_{\delta y \rightarrow 0} P(\{X \leq x\} | y \leq Y \leq y + \delta y)$$

Let $f_{X|Y}(x|y)$ be pdf st $F_{X|Y}(x|y) = \int_{-\infty}^x f_{X|Y}(u|y) du$

We want to show that

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$F_{X|Y}(x|y) = \lim_{\delta_y \rightarrow 0} \frac{P(\{X \leq x\}, y \leq Y \leq y + \delta_y)}{P(y \leq Y \leq y + \delta_y)}$$

$$= \lim_{\delta_y \rightarrow 0} \frac{\int_y^{y+\delta_y} \int_{-\infty}^x f_{X,Y}(u,v) du dv}{\int_y^{y+\delta_y} f_Y(v) dv}$$

$$\begin{aligned} f_{X,Y}(u,v) &\approx f_{X,Y}(u,y) \quad \forall v \in [y, y+\delta_y] \\ f_Y(v) &\approx f_Y(y) \quad \text{"} \end{aligned}$$

$$= \lim_{\delta y \rightarrow 0} \frac{\int_{-\infty}^x f_{X,Y}(u,y) \delta y \, du}{f_Y(y) \delta y}$$

$$F_{X|Y}(x|y) = \int_{-\infty}^x \boxed{\frac{f_{X,Y}(u,y)}{f_Y(y)}} \, du$$

" $f_{X|Y}(x|y)$

Conditional pdf's - X, Y continuous.

$$① f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$② f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$$

Multiplication

$$f_{X,Y}(x,y)$$

Example

$$f_{X,Y}(x,y)$$

$$f_X(x)$$

$$(u, y) \otimes_y du$$

$$g) \otimes_y$$

$$\frac{(u, y)}{(y)} du$$

$$r(x|y)$$

$$x, y \text{ continuous}$$

$$x, y)$$

$$r(y)$$

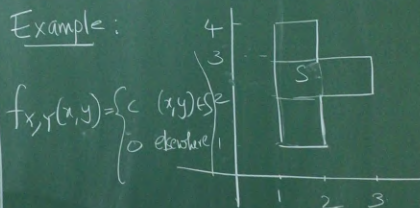
$$r(x, y)$$

$$f_{X,Y}$$

Multiplicative rule

$$\begin{aligned} f_{X,Y}(x, y) &= f_X(x) f_{Y|X}(y|x) \\ &= f_Y(y) f_{X|Y}(x|y) \end{aligned}$$

Example:



$$c \times 4 = 1 \Rightarrow c = \frac{1}{4}$$

$$f_X(x) = \begin{cases} \frac{1}{4} & x=1, 2 \\ 0 & \text{elsewhere} \end{cases} \quad f_Y(y) = \begin{cases} \frac{1}{2} & y=3, 4 \\ 0 & \text{elsewhere} \end{cases}$$

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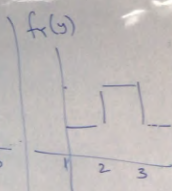
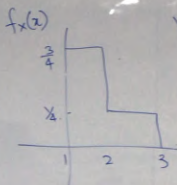
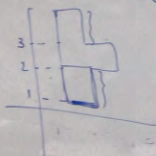
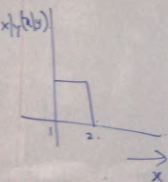
$$f_X(x) = \begin{cases} 3/4 & x \in [1, 2] \\ 1/4 & x \in [2, 3] \\ 0 & \text{otherwise} \end{cases}$$

$$f_Y(y) = \begin{cases} 1/4 & y \in [1, 2] \cup [3, 4] \\ 1/2 & y \in [2, 3] \\ 0 & \text{otherwise} \end{cases}$$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$= \begin{cases} \frac{c}{1/4} = 1 & y \in [1, 2] \cup [3, 4] \\ \frac{c}{1/2} = 1/2 & y \in [2, 3] \end{cases}$$

$$y \in [1, 2] \cup [3, 4]$$



$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx$$

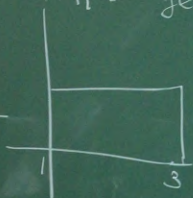
$$= \begin{cases} \int_1^2 c dx = 1/2 & y \in [2, 3] \\ \int_1^2 c dx = 1/4 & y \in [1, 2] \cup [3, 4] \end{cases}$$

Conditional pdfs

$$\int_{-\infty}^{\infty} f_{X|Y}(x|y) dx = \int_{-\infty}^{\infty} \frac{f_{X,Y}(x,y)}{f_Y(y)} dx.$$

$$f_{X|Y}(y) \quad y \in [2,5]$$

$$= \frac{f_Y(y)}{f_Y(b)} = 1$$



$f_{X|Y}(y|a)$ Exercise

Example 2:

$$f_{X,Y}(x,y) = \begin{cases} c & x^2 + y^2 \leq r^2 \\ 0 & \text{otherwise} \end{cases}$$

find $f_X(x)$, $f_Y(y)$, $f_{X|Y}(x|y)$, $f_{Y|X}(y|x)$.

$$\int_{-r}^r \int_{-\sqrt{r^2-y^2}}^{\sqrt{r^2-y^2}} c \, dx \, dy = 1$$

$$c \times \pi r^2 = 1$$

$$c = \frac{1}{\pi r^2}$$



$y(x)$.



$$f_x(x) = \int_{-\infty}^{\infty} f_{x,y}(x,y) dy$$

$$= \int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}} c dy$$

$$= \frac{2x\sqrt{r^2-x^2}}{\pi r^2}$$

$$f_y(y) = \frac{2\sqrt{r^2-y^2}}{\pi r^2}$$

$$f_{x,y}(x,y)$$

$$= \left\{ \begin{array}{l} \dots \end{array} \right.$$

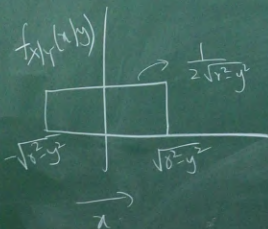
$$f_{x,y}(x,y)$$

$$\sqrt{r^2-y^2}$$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$= \begin{cases} \frac{\frac{1}{\sqrt{r^2-y^2}}}{2\sqrt{r^2-y^2}} = \frac{1}{2\sqrt{r^2-y^2}} & -\sqrt{r^2-y^2} \leq x \leq \sqrt{r^2-y^2} \\ 0 & \text{otherwise} \end{cases}$$

$$X|Y=y \sim \text{Uniform}(-\sqrt{r^2-y^2}, \sqrt{r^2-y^2})$$



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X discrete and Y is continuous.

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

Can't describe joint PMF or joint pdf in this case.

Conditional PMF $P_{X|Y}(x|y)$, $f_Y(y)$ or

Conditional pdf $f_{Y|X}(y|x)$, $P_X(x)$.

$$P_{X|Y}(x|y) = \lim_{\delta_y \rightarrow 0} P(\{X=x\} | \{y \leq Y \leq y + \delta_y\})$$

$$F_{Y|X}(y|x) = P(Y \leq y | X=x)$$

$$f_{Y|X}(y|x) \text{ pdf st } F_{Y|X}(y|x) = \int_{-\infty}^y f_{Y|X}(v|x) dv$$

Multiplication rule for X discrete, Y continuous.

$$P_{X|Y}(x|y) f_Y(y) = f_{Y|X}(y|x) P_X(x).$$

$$P(X=x, y \leq Y \leq y+\delta y).$$

$$= P(X=x) P(y \leq Y \leq y+\delta y | X=x)$$

$$= P(y \leq Y \leq y+\delta y) P(X=x | y \leq Y \leq y+\delta y)$$

$$\rightarrow = P(X=x) \int_y^{y+\delta y} f_{Y|X}(v|x) dv.$$

$$\approx P(X=x) f_{Y|X}(y|x) \delta y.$$

$$= \int_y^{y+\delta y} f_Y(v) dv \cdot P_{X|Y}(x|y \leq Y \leq y+\delta y)$$

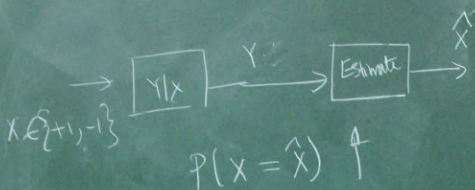
$$= f_Y(y) \delta y \cdot P_{X|Y}(x|y \leq Y \leq y+\delta y)$$

$$\Rightarrow P(X=x) f_{Y|X}(y|x) = f_Y(y) P_{X|Y}(x|y \leq Y \leq y+\delta y)$$

lim on both LHS & RHS.
 $\delta y \rightarrow 0$

$$P(X=x) f_{Y|X}(y|x) = f_Y(y) P_{X|Y}(x|y)$$

Bayes rule:



Data set has images of handwritten

16

2

3

3

2

7

7

9

9

← 16 →

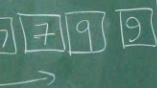


Distribution of X is referred to as "prior" distribution

$Y|X \rightarrow$ "Likelihood"

geschrieben

"Posterior distribution"



$$P(X=x | Y=y)$$

Maximum A posterior Probability estimator (MAP).

$$\hat{x} = \hat{x}(y) \text{ and } \hat{x}(y) = \underset{x}{\operatorname{argmax}} P(X=x | Y=y)$$